CONCAVITY OF THE ENTROPY OF DETERMINANTAL MEASURES ON TWO POINTS, AND THE SECOND VARIATION IN GENERAL

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Abstract. For a positive contraction Q on $\ell^2(E)$ with E finite, let P_Q be the determinantal probability measure on 2^E and $\mathrm{Ent}(Q)$ its Shannon entropy. Lyons conjectured in 2003 that Ent is concave in Q. We prove the conjecture when $|E| = 2$. For general E we write the second derivative of Ent along a line as a negative quadratic form in the derivatives of the principal minors of Q plus a sum, over subsets T with $|T| \geq 2$, of the second derivatives of the minors $\det Q_T$ weighted by Möbius sums of $\log P_Q$. For two-element T these weights are nonnegative by the Hadamard–Fischer inequality, which is what makes the case $|E| = 2$ work. We also report numerical evidence for $|E| = 3, 4$: the second derivative is negative away from scalar kernels cI, where it vanishes in the off-diagonal directions.

1. Introduction

Let E be a finite set and Q a positive contraction on $\ell^2(E)$, that is, a Hermitian matrix with $0 \leq Q \leq I$. The determinantal probability measure P_Q on 2^E is determined by $P_Q[\, A \subseteq Y \,] = \det Q_A$ for $A \subseteq E$, where Q_A is the principal submatrix indexed by A [1]. Its entropy is

$$\mathrm{Ent}(Q) \;=\; -\sum_{S \subseteq E} P_Q[Y = S] \log P_Q[Y = S].$$

In Section 9 of [1], Lyons asked whether Ent is concave on the convex set $\{0 \leq Q \leq I\}$ and stated the following conjecture, supported by numerical calculation.

Conjecture 1.1 (Lyons [1, Conjecture 9.2])**.** *For positive contractions Q_1, Q_2 on $\ell^2(E)$,* $\mathrm{Ent}\big((Q_1 + Q_2)/2\big) \geq \big(\mathrm{Ent}(Q_1) + \mathrm{Ent}(Q_2)\big)/2$.

The conjecture is restated as Conjecture 2.6 of Lyons's survey [2]. By finite restrictions it implies Conjecture 5.3 there: the entropy of the stationary determinantal process P_f with symbol f, studied by Lyons and Steif [3], is a concave function of f. The map $Q \mapsto P_Q$ is polynomial of degree $|E|$, so concavity of Ent is not inherited from the concavity of $x \mapsto -x \log x$.

Theorem 1.2. *Conjecture 1.1 holds when $|E| = 2$.*

Section 2 proves Theorem 1.2. Section 3 gives the second variation of Ent for arbitrary finite E (Proposition 3.1) and explains which part of the two-point argument survives. Section 4 records numerical evidence for $|E| = 3$ and 4.

2. Two points

Let $E = \{1, 2\}$ and write $Q = \left(\begin{smallmatrix} a & b \\ \bar{b} & d \end{smallmatrix}\right)$. The four probabilities are

$$p_{12} = \det Q, \qquad p_1 = a - \det Q, \qquad p_2 = d - \det Q, \qquad p_0 = 1 - a - d + \det Q, \tag{2.1}$$

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where $p_S = P_Q[Y = S]$ and $p_0 = P_Q[Y = \emptyset]$. A direct computation gives

$$p_1 p_2 - p_0 p_{12} = |b|^2. \tag{2.2}$$

Since Ent is continuous on the compact set $\{0 \le Q \le I\}$, it suffices to prove concavity on the open set $\{0 < Q < I\}$, where all four probabilities are positive: $p_0 = \det(I - Q) > 0$ and $p_{12} = \det Q > 0$, so $p_1 p_2 > 0$ by (2.2), and $p_1 + p_2 = \lambda_1 + \lambda_2 - 2\lambda_1\lambda_2 > 0$ for the eigenvalues $\lambda_1, \lambda_2 \in (0, 1)$ of Q. On that set concavity is equivalent to $f''(0) \le 0$ for every Q and every Hermitian D, where $f(t) = \mathrm{Ent}(Q + tD)$.

Put $\varphi(x) = -x \log x$ and

$$G(a, d, g) = \varphi(g) + \varphi(a - g) + \varphi(d - g) + \varphi(1 - a - d + g),$$

so that $\mathrm{Ent}(Q) = G(a, d, \det Q)$ by (2.1). The function G is a sum of concave functions of affine maps, hence concave, and

$$\frac{\partial G}{\partial g} = \log \frac{p_1 p_2}{p_0 p_{12}} =: L \ \ge 0 \tag{2.3}$$

by (2.2). Along $Q + tD$ the entries a, d move affinely, $\frac{d}{dt} \det(Q + tD)\big|_0 = \mathrm{tr}(\mathrm{adj}(Q)D) =: \gamma$ and $\frac{d^2}{dt^2} \det(Q + tD) = 2 \det D$. Writing $x = D_{11}$, $y = D_{22}$, the chain rule gives

$$f''(0) = -S + 2L \det D, \qquad S = \frac{\ell_{12}^2}{p_{12}} + \frac{\ell_1^2}{p_1} + \frac{\ell_2^2}{p_2} + \frac{\ell_0^2}{p_0}, \tag{2.4}$$

with $\ell_{12} = \gamma$, $\ell_1 = x - \gamma$, $\ell_2 = y - \gamma$, $\ell_0 = \gamma - x - y$. One checks directly that

$$\ell_1 \ell_2 - \ell_{12} \ell_0 = xy. \tag{2.5}$$

Lemma 2.1. *For $0 < Q < I$ on $\mathbb{C}^2$, the matrix $\mathrm{adj}\, Q$ has the same eigenvalues as Q; in particular $0 \le \mathrm{adj}\, Q \le I$.*

Proof. $\mathrm{adj}\, Q = (\det Q)\, Q^{-1}$ has eigenvalues $\lambda_1\lambda_2/\lambda_1 = \lambda_2$ and $\lambda_1\lambda_2/\lambda_2 = \lambda_1$. □

Lemma 2.2. *Let $0 < Q < I$ and $D \ge 0$ on $\mathbb{C}^2$, and put $A = \gamma\,(\mathrm{tr}\, D - \gamma) = -\ell_{12}\ell_0$ and $r = \sqrt{p_0 p_{12}}$. Then $A \ge (p_1 + p_2 + 2r) \det D$.*

Proof. Work in an orthonormal eigenbasis of $\mathrm{adj}\, Q$, with eigenvalues $\mu_1 = \lambda_2$, $\mu_2 = \lambda_1$ (Lemma 2.1), and let $\alpha, \delta \ge 0$ be the diagonal entries of D in that basis, so $\alpha\delta \ge \det D$. Then $\gamma = \mu_1\alpha + \mu_2\delta$ and $\mathrm{tr}\, D - \gamma = (1 - \mu_1)\alpha + (1 - \mu_2)\delta$, and

$$A = \mu_1(1 - \mu_1)\alpha^2 + \mu_2(1 - \mu_2)\delta^2 + \big[\mu_1(1 - \mu_2) + \mu_2(1 - \mu_1)\big]\alpha\delta.$$

The bracket equals $\lambda_1 + \lambda_2 - 2\lambda_1\lambda_2 = p_1 + p_2$, and by the arithmetic–geometric mean inequality the first two terms are at least $2\sqrt{\mu_1\mu_2(1 - \mu_1)(1 - \mu_2)}\,\alpha\delta = 2r\,\alpha\delta$, because $\lambda_1\lambda_2 = p_{12}$ and $(1 - \lambda_1)(1 - \lambda_2) = p_0$. Hence $A \ge (p_1 + p_2 + 2r)\alpha\delta \ge (p_1 + p_2 + 2r) \det D$. □

Proof of Theorem 1.2. Fix $0 < Q < I$ and a Hermitian D. If $\det D \le 0$ then $f''(0) \le 0$ by (2.4) and (2.3). Otherwise D is definite, and since (2.4) is even in D we may assume $D \ge 0$. Put $s = \sqrt{p_1 p_2}$ and $r = \sqrt{p_0 p_{12}}$, so $s \ge r$ by (2.2). Applying $u^2/p + v^2/q \ge 2|uv|/\sqrt{pq}$ to the pairs (ℓ_{12}, ℓ_0) and (ℓ_1, ℓ_2),

$$S \ \ge \ \frac{2|\ell_{12}\ell_0|}{r} + \frac{2|\ell_1\ell_2|}{s} = \frac{2(|A|s + |B|r)}{rs}, \qquad B := \ell_1\ell_2.$$

The logarithmic mean of s^2 and r^2 is at least their geometric mean, so $L = \log(s^2/r^2) \le (s^2 - r^2)/(rs)$. By (2.4) it therefore suffices to show

$$|A|s + |B|r \ \ge \ (s^2 - r^2) \det D. \tag{2.6}$$

By Lemma 2.1 and $D \geq 0$ we have $\gamma \geq 0$ and $\operatorname{tr} D - \gamma = \operatorname{tr}((I - \operatorname{adj} Q)D) \geq 0$, so $A \geq 0$; and $A + B = xy \geq \det D \geq 0$ by (2.5).

If $B < 0$, then $A > xy \geq \det D$, and $s \leq \frac{1}{2}(p_1 + p_2) \leq 1$ gives $As \geq s^2 \det D \geq (s^2 - r^2) \det D$.

If $B \geq 0$, then $As + Br = (s-r)A + r\,xy$. Lemma 2.2 and $p_1 + p_2 \geq 2s$ give $A \geq (2s+2r) \det D \geq (s + r) \det D$, hence $As + Br \geq (s - r)(s + r) \det D + r\,xy \geq (s^2 - r^2) \det D$.

In both cases (2.6) holds, so $f''(0) \leq 0$. Thus Ent is concave on $\{0 < Q < I\}$ and, by continuity, on $\{0 \leq Q \leq I\}$. □

Remark 2.3. The bound is far from tight in the middle of the parameter range. Over 2×10^5 random pairs (Q, D) with D definite, the ratio $2L \det D/S$ never exceeded 0.28. Equality $f''(0) = 0$ occurs, for instance, at $b = 0$, $a = d$ (that is, $Q = cI$) with D off-diagonal: there P_Q is a product measure, and moving Q off the diagonal keeps the one-point marginals fixed while P_Q changes only at second order, so the entropy, which is maximal among laws with those marginals, changes at fourth order.

3. The second variation for general E

For $T \subseteq E$ write $m_T = \det Q_T$ (with $m_\emptyset = 1$). Inclusion–exclusion gives

$$P_Q[Y = S] = \sum_{T \supseteq S} (-1)^{|T \setminus S|} m_T, \tag{3.1}$$

so $P_Q[Y = \cdot\,]$ is an affine function of the vector of principal minors. Differentiating $\mathrm{Ent} = \sum_S \varphi(P_Q[Y = S])$ twice along $Q + tD$ gives the following.

Proposition 3.1. *Let $0 < Q < I$, D Hermitian, $p_S = P_Q[Y = S]$, and m'_T, m''_T the first two derivatives of $\det(Q + tD)_T$ at $t = 0$. Then*

$$\frac{d^2}{dt^2} \operatorname{Ent}(Q + tD)\Big|_{t=0} = -\sum_{S \subseteq E} \frac{1}{p_S} \Big(\sum_{T \supseteq S} (-1)^{|T \setminus S|} m'_T \Big)^2 + \sum_{|T| \geq 2} \Lambda_T\, m''_T,$$

where $\Lambda_T = -\sum_{S \subseteq T} (-1)^{|T \setminus S|} \log p_S$.

Proof. By (3.1), $\frac{d}{dt} p_S = \sum_{T \supseteq S} (-1)^{|T \setminus S|} m'_T$ and $\frac{d^2}{dt^2} p_S = \sum_{T \supseteq S} (-1)^{|T \setminus S|} m''_T$. Since $\varphi'(x) = -1 - \log x$ and $\varphi''(x) = -1/x$,

$$\frac{d^2}{dt^2} \operatorname{Ent} = \sum_S \varphi''(p_S) \Big(\frac{dp_S}{dt} \Big)^2 + \sum_T m''_T \sum_{S \subseteq T} (-1)^{|T \setminus S|} \varphi'(p_S).$$

The constant -1 in φ' contributes $\sum_{S \subseteq T} (-1)^{|T \setminus S|} = 0$ for $T \neq \emptyset$, and $m''_T = 0$ for $|T| \leq 1$ because m_T is then affine in t. □

When $Q < I$, the measure P_Q is the L-ensemble with $L = Q(I - Q)^{-1}$, that is $p_S = \det L_S / \det(I + L)$. The normalisation cancels in Λ_T for $T \neq \emptyset$, so

$$\Lambda_T = -\sum_{S \subseteq T} (-1)^{|T \setminus S|} \log \det L_S.$$

For $T = \{i, j\}$,

$$\Lambda_{ij} = \log \frac{L_{ii} L_{jj}}{\det L_{\{i,j\}}} \geq 0$$

by the Hadamard–Fischer inequality; for $|E| = 2$ this is (2.3). The proof of Theorem 1.2 uses exactly two features of the case $|E| = 2$: the only curvature term is $\Lambda_{12}\, m''_{12} = 2\Lambda_{12} \det D$, with $\Lambda_{12} \geq 0$, and Λ_{12} is controlled by the quadratic form through the logarithmic-mean inequality. For $|E| \geq 3$ the coefficients Λ_T with $|T| \geq 3$ take both signs: for $|E| = 3$, in 3000 random kernels

with spectra in $[0.05, 0.95]$, Λ_{123} was negative in 1508 cases, and $\Lambda_T m_T'' > 0$ occurred for both $|T| = 2$ and $|T| = 3$. The curvature terms therefore have to be compared with the quadratic form jointly.

4. Numerical evidence for three and four points

We searched for counterexamples in two ways, with kernels $Q = U \operatorname{diag}(\lambda) U^*$ for U from the QR factorisation of a Gaussian matrix (real or complex) and $\lambda \in (0,1)^n$.

Midpoint gaps. For $n = |E| = 3$ (real) and $n = 4$ (complex) we minimised $\operatorname{Ent}\big((Q_1+Q_2)/2\big) - \big(\operatorname{Ent}(Q_1) + \operatorname{Ent}(Q_2)\big)/2$ by Nelder–Mead from the worst of 2000 and 3000 random pairs. The minima were 8×10^{-15} and 2×10^{-8}, attained as $Q_1 - Q_2 \to 0$.

Second derivatives. We maximised the central difference $\big(\operatorname{Ent}(Q + hD) + \operatorname{Ent}(Q - hD) - 2\operatorname{Ent}(Q)\big)/h^2$ with $h = 10^{-3}$ and $\|D\|_F = 1$, keeping the spectrum of Q at distance at least $5h$ from $\{0, 1\}$. For $n = 3$ (real) and $n = 4$ (complex) the maxima were -2×10^{-6} and -1.2×10^{-6}, approached only near scalar kernels $Q = cI$ (in every optimised kernel the eigenvalues agreed to within 4×10^{-4}). Near those points, for $n = 3$, $Q = cI + \varepsilon A$ with $c \in [0.2, 0.8]$, ε down to 3×10^{-3} and 400 random real symmetric pairs (A, D), half of them with D off-diagonal, the second derivative stayed negative. This matches Remark 2.3: scalar kernels give product measures, and off-diagonal perturbations preserve the one-point marginals.

The formula of Proposition 3.1 agrees with central differences to relative accuracy 3×10^{-6} at $n = 3$.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

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Data availability

No data were used. The code for the computations of Remark 2.3 and Sections 3–4, with their outputs, is available at
`https://github.com/yspennstate/lyons-determinantal-entropy-two-points`.

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